CocoConnect

Requirement Analysis Document

Version 2 – Updated after module amendments and feasibility review

Client: Coconut Growers Association of Sri Lanka · Group 38

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Document History

Version	Content
1	Initial requirement analysis: stakeholders, current problems, domain knowledge, consolidated and prioritised requirements, gaps between sources
2	This update: buyer and estate owner module amendments, requirements revisited with stakeholders, changes driven by the technical feasibility findings, resolution of the open issues from version 1, and the refined development approach

Related documents: Functional Analysis Document v2, Technical Feasibility Document v1 and v2, SRS Version 3.

1. Purpose

This document updates the requirement analysis after the buyer and estate owner modules were amended and the requirements were revisited with stakeholders. It records what changed and why, closes the open issues from version 1, applies the findings of the technical feasibility study, and sets out the refined development approach used for implementation.

2. Summary of Changes

- Coconut products organised by husk condition: raw, husked and dehusked.
- Price agreement moved into an in-app negotiation chat with price offers.
- Buyer requests and estate listings are matched automatically by quantity and date.
- Location captured at sign-up; marketplaces filtered by distance.
- Deals are tracked in an Orders hub and completed by the estate owner, with admin review of price changes.
- Service marketplace extended with a map of nearby machinery and a service calendar.
- Online payment and offline harvest recording scheduled for later, based on feasibility.

3. Buyer Module Amendments

Area	Version 1	Version 2	Change
Product categories	DC coconut, curry coconut, husk	Buying Categories, then coconut type by husk condition: raw, husked, dehusked	**Changed**
Selling unit	Per kilogram and per nut	Per nut first; per kilogram later	**Changed**
Finding supply	Browse, search and filter	Nearby listings with 10 / 25 / 50 km filters and price sorting	**Changed**
Buy requests	Not defined	Post a request with quantity, required date and bidding price	**New**
Matching	Not defined	Listings with at least the requested quantity, available on the required date or up to 7 days after	**New**
Contacting sellers	Contact sellers	Negotiation chat with price offers; accept or reject the estate's offer	**Changed**
Tracking	Not defined	Orders hub: Pending Orders, Deals with due dates, Transactions	**New**
Alerts	Notifications	Alerts for offers and new messages with badge counts	**Changed**

4. Estate Owner Module Amendments

Area	Version 1	Version 2	Change
Dashboard	Not defined	Selling categories and provider services on one dashboard	**New**
Selling	Create listings	Sell Summary: quantity, available date and price per nut, then post	**Changed**
Finding buyers	Not defined	Interested Buyers: matching requests sorted by best price, earliest date or nearest	**New**

Area	Version 1	Version 2	Change
Responding to buyers	Not defined	Send an offer on a buyer request, which opens a chat	**New**
Negotiation	Not defined	Accept or reject the buyer's counter-offers in chat	**New**
Completing deals	Not defined	Record delivered quantity and final price; a different price needs admin review	**New**
Market information	Not defined	Latest closed deal and average price per nut	**New**
Services	Find workers	Map of nearby machinery with daily rates; service calendar	**Changed**

5. Changes from the Feasibility Findings

Finding	Effect on requirements
Google Maps needs a billing account	Maps use OpenStreetMap; location is picked on a map at sign-up
Online payment has low feasibility	Payment stays outside the app; the system records the agreed price and total
Offline sync is medium feasibility	Offline harvest recording is built after the marketplaces
Negotiation must stay consistent for both sides	Only the receiving side may accept an offer; a new offer replaces the pending one; accepting creates the order in one transaction
Service calendar risks double bookings	A provider cannot be booked for overlapping dates
Marketplace parts can be reused	Service marketplace requirements reuse distance filters, request and accept flow and alerts

6. Updated Requirements

6.1 User management

ID	Requirement	Priority	Status
UM-01	Register with name, phone, password, role and location picked on a map	**Must**	**Changed**
UM-02	Log in with phone number and password; stay logged in	**Must**	**Changed**
UM-03	Role-based access for buyer, estate owner, service provider and admin	**Must**	**Unchanged**
UM-04	Update profile and location	**Must**	**Unchanged**
UM-05	Reset password	**Should**	**Unchanged**

6.2 Product marketplace

ID	Requirement	Priority	Status
PM-01	Estate owner posts a sell offer: coconut type, quantity, available date, price per nut	**Must**	**Changed**
PM-02	Buyer browses listings by coconut type with distance	**Must**	**Changed**
PM-03	Filter by distance (10 / 25 / 50 km) and sort by price	**Must**	**Changed**
PM-04	Buyer posts a request: quantity, required date, bidding price	**Must**	**New**
PM-05	Automatic matching by quantity and a 7-day date window, for both sides	**Must**	**New**
PM-06	Estate owner sends one offer per buyer request	**Must**	**New**
PM-07	Selling per kilogram	**Could**	**Deferred**

6.3 Negotiation and orders

ID	Requirement	Priority	Status
NO-01	Chat between buyer and estate owner with messages and price offers	**Must**	**New**
NO-02	Only the receiving side accepts or rejects an offer; a new offer replaces the pending one	**Must**	**New**
NO-03	Accepting an offer creates an order and closes other offers on the same request	**Must**	**New**

ID	Requirement	Priority	Status
NO-04	Orders hub with Pending Orders, Deals (due dates, filters) and Transactions	**Must**	**New**
NO-05	Estate owner completes a deal with delivered quantity and final price	**Must**	**New**
NO-06	A final price different from the agreed price is reviewed by an admin	**Must**	**New**
NO-07	Market price: latest deal and average price per nut	**Should**	**New**
NO-08	Alerts for offers and unread messages	**Must**	**Changed**
NO-09	Online payment	**Future**	**Deferred**

6.4 Service marketplace

ID	Requirement	Priority	Status
SM-01	Service profiles with skill type, charging method and rate	**Must**	**Unchanged**
SM-02	Machinery listed with daily rate	**Must**	**Changed**
SM-03	Map of nearby machinery with price tags	**Should**	**New**
SM-04	Hiring requests with accept or reject and job status	**Must**	**Unchanged**
SM-05	Service calendar of confirmed and pending bookings, without double bookings	**Should**	**New**

6.5 Other modules

ID	Requirement	Priority	Status
OT-01	Ratings and reviews after completed orders and jobs	**Should**	**Unchanged**
OT-02	Estate registration and estate information	**Must**	**Unchanged**
OT-03	Harvest recording, including offline	**Should**	**Changed**
OT-04	Admin: manage users, listings, reports and price disputes	**Must**	**Changed**
OT-05	English, Sinhala and Tamil interface	**Must**	**Changed**

7. Open Issues from Version 1

Version 1 issue	Decision	Status
Multilingual support listed as future in the SRS draft	Core requirement: EN | SI | TA switch on main screens	**Resolved**
Location services listed as future in the SRS draft	In scope: location at sign-up and distance filters	**Resolved**
Offline use: SRS assumes internet, stakeholders need offline harvest	Only harvest recording works offline	**Resolved**
Ten stakeholder roles vs six user roles	Six system roles; worker types become service categories	**Resolved**
Equipment rental: current or future	Machinery with daily rates now; full rental booking later	**Resolved**
How a deal is confirmed and what happens if it fails	Accepted offer creates an order; estate owner completes it; price changes go to admin review	**Resolved**

8. Refined Development Approach

Practice	How it is applied
Agile sprints	Short sprints, each ending with working features shown to the client and supervisors
Build order	Users and location → listings and requests → negotiation and orders → alerts → service marketplace → ratings → offline harvest
API first	Each API is documented in Swagger before the screens are connected
Branch per member	Each member works on their own GitHub branch; work is merged into an integration branch
Shared database	One cloud PostgreSQL database; changes made with numbered SQL scripts that are safe to re-run

Practice	How it is applied
Testing	Automated backend tests on an in-memory database; manual testing on the Android emulator
Definition of done	Works on the emulator against the real backend, tests pass, documented in Swagger, merged

9. Traceability: Problems to Updated Requirements

Stakeholder problem	Updated requirements
No central marketplace; limited visibility of stock	PM-01 to PM-05
Prices negotiated manually by phone	NO-01 to NO-03
Price agreed before husking	NO-02, NO-03
Trust between buyers and sellers	NO-06, OT-01, UM-03
Hard to find workers and machinery	SM-01 to SM-05
No price transparency	NO-07, PM-03
Manual harvest records	OT-03
Users speak different languages	OT-05